THE CHINESE UNIVERSITY OF HONG KONG, SHENZHEN
School of Management and Economics

ECONOMICS 3121
Introductory Econometrics

Fall term 2015 **MIDTERM EXAM** Yi Ding

DATE: **Wednesday November 4, 2015**

TIME: **120 minutes; 8:30 a.m. – 10:30 a.m.**

INSTRUCTIONS: The exam consists of **FOUR (4)** questions. Students are required to answer **ALL FOUR (4)** questions.

Answer all questions in the exam booklets provided. Be sure your ***name*** and ***student number*** are printed clearly on the front of all exam booklets used.

Do not write answers to questions on the front page of the first exam booklet.

Please label clearly each of your answers in the exam booklets with the appropriate number and letter.

Please write legibly.

This exam is **CONFIDENTIAL**. This question paper must be submitted in its entirety with your answer booklet(s); otherwise your exam will not be marked.

A formula sheet and tables of critical values of the t distribution and F distribution are given on the last pages of the exam.

MARKING: The marks for each question are indicated in parentheses immediately above each question. **Total marks** for the exam **equal 110**.

GOOD LUCK!

Question 1: Multiple Choice (40 points, 2 points each)
Select the **BEST** response to each of the following questions.

1 . Which of the following is true?
a) A variable has a causal effect on another variable if both variables increase or decrease simultaneously.
b) The notion of 'ceteris paribus' plays an important role in causal analysis.
c) Difficulty in inferring causality disappears when studying data at fairly high levels of aggregation.
d) The problem of inferring causality arises if experimental data is used for analysis.

Answer: b

2. If $\widehat{\beta_j}$, an unbiased estimator of β_j, is consistent, then the:
a) distribution of $\hat{\beta}_j$ becomes more and more loosely distributed around β_j as the sample size grows.
b) distribution of $\hat{\beta}_j$becomes more and more tightly distributed around β_j as the sample size grows.
c) distribution of $\hat{\beta}_j$ tends toward a standard normal distribution as the sample size grows.
d) distribution of $\hat{\beta}_j$ remains unaffected as the sample size grows.

Answer: b

3. Which of the following statements is true?
a) If the calculated value of F statistic is higher than the critical value, we reject the alternative hypothesis in favor of the null hypothesis.
b) The F statistic is always nonnegative as SSR_r is never smaller than SSR_{ur}.
c) Degrees of freedom of a restricted model is always less than the degrees of freedom of an unrestricted model.
d) The F statistic is more flexible than the t statistic to test a hypothesis with a single restriction.

Answer: b

4. Which of the following is a nonlinear regression model?
a) $y = \beta_0 + \beta_1 x^{1/2} + u$
b) $\log(y) = \beta_0 + \beta_1 \log(x) + u$
c) $y = 1/(\beta_0 + \beta_1 x) + u$
d) $y = \beta_0 + \beta_1 x + u$

Answer: c

5. Consider the following regression model: $y = \beta_0 + \beta_1 x + u$. Which of the following is a property of Ordinary Least Square (OLS) estimates of this model and their associated statistics?
a) The sum, and therefore the sample average of the OLS residuals, is positive.
b) The sum of the OLS residuals is negative.
c) The sample covariance between the regressors and the OLS residuals is positive.
d) The point $(\bar{x}, \bar{y})$ always lies on the OLS regression line.

Answer: d

6. Exclusion of a relevant variable from a multiple linear regression model leads to the problem of _____.

a) misspecification of the model
b) multicollinearity
c) perfect collinearity
d) homoskedasticity

Answer: a

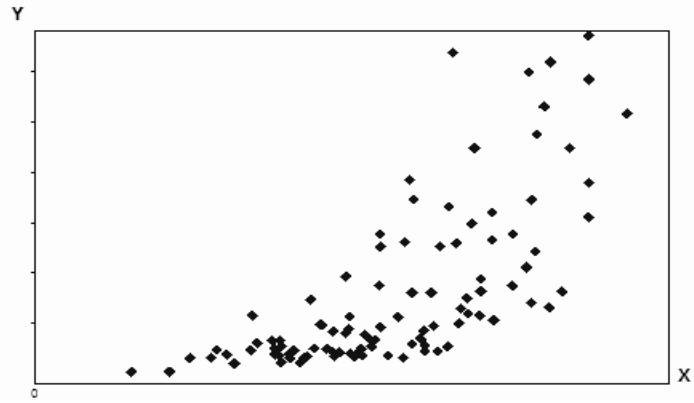
7. Suppose the variable x_2 has been omitted from the following regression equation, $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$. $\widetilde{\beta_1}$ is the estimator obtained when x_2 is omitted from the equation. The bias in $\widetilde{\beta_1}$ is positive if _____.

a) $\beta_2 > 0$ and x_1 and x_2 are positively correlated
b) $\beta_2 < 0$ and x_1 and x_2 are positively correlated
c) $\beta_2 > 0$ and x_1 and x_2 are negatively correlated
d) $\beta_2 = 0$ and x_1 and x_2 are negatively correlated

Answer: a

8. Which regression model would best fit the data shown in the plot below?

a) $y = \beta_0 + \beta_1 x + u$
b) $y = \beta_0 + \beta_1 \log(x) + u$
c) $\log(y) = \beta_0 + \beta_1 x + u$
d) $\log(y) = \beta_0 + \beta_1 \log(x) + u$



Answer: c

9. A researcher has just finished a statistical analysis and claims that he found evidence that x affects y positively. Using a 1-tailed test, he found that the estimated coefficient for x is significantly positive at the 5% level, but is not significant at the 4% level. Assume that the classical linear regression assumptions hold. If the researcher used a 2-tailed test instead, what would he have found?

a) The estimated coefficient for x is significantly positive at the 10% level.
b) The estimated coefficient for x is significantly positive at the 5% level.
c) The estimated coefficient for x is significantly positive at the 4% level.
d) The estimated coefficient for x is significantly positive at the 2.5% level.

Answer: a

10. Which of the following statements is true when the dependent variable, > 0 ?
    a) Taking log of a variable often expands its range.
    b) Models using $log(y)$ as the dependent variable will satisfy CLM assumptions more closely than models using the level of y.
    c) Taking log of variables make OLS estimates more sensitive to extreme values.
    d) Taking logarithmic form of variables make the slope coefficients more responsive to rescaling.

Answer: b

11. The Gauss-Markov theorem shows that, for linear regression models, the least squares estimator for the slope coefficients is *unbiased*. That the slope estimator is *unbiased* means that:
    a) the estimated slope coefficient will always be equal to the true parameter value
    b) the estimated slope coefficient will get closer to the true parameter value as the size of the sample increases
    c) the estimated slope coefficient will be equal to the true parameter if the sample is large
    d) if repeated samples of the population are taken, on average the value of the slope estimator will be equal to the true parameter value

Answer: d

12. In a multiple OLS regression, if you divide the explained variable y by 12 (to change units of measurement from inches to feet) which of the following will happen?
    a) The estimated intercept parameter will be reduced by 12 times.
    b) the estimated slope parameter will be reduced by 12 units.
    c) the total sum of squares for the regression will be reduced by $\sqrt{12}$ times
    d) R-squared for the regression will be reduced by 12 times.

Answer: a

13. In a multiple OLS regression, what is the consequence of including an irrelevant variable?
    a) The OLS estimators are biased.
    b) The R-square of the regression may decrease.
    c) The t statistic the OLS estimators may increase
    d) The estimated variance of the OLS estimators may increase.

Answer: d

14. You have just read a paper which claims that there is strong evidence that x affects y negatively. The researcher found that the estimated coefficient for x is significantly negative at the 1% level. You suspect that there is omitted variable bias. Which of the following statements is correct?

    a) If there is omitted variable bias, and the bias is positive, then the researcher's conclusions are invalid.
    b) If there is omitted variable bias, and the bias is positive, then the researcher's conclusions are still valid.
    c) If there is omitted variable bias, and the bias is zero, then the researcher's conclusions are invalid.

d) If there is omitted variable bias, and the bias is negative, then the researcher's conclusions are still valid.

Answer: a

You have been commissioned to construct a model that explains NBA players' wages. The dependent variable is *lwage*, the natural log of players' annual wages (*in thousands of dollors*). Please use the following Stata regression results for the next **SIX** questions.

Source	SS	df	MS	Number of obs =	269
				F(5, 263) =	40.27
Model	90.2698185	5	18.0539637	Prob > F =	0.0000
Residual	117.918945	263	.448361006	R-squared =	0.4336
				Adj R-squared =	0.4228
Total	208.188763	268	.776823743	Root MSE =	.6696

lwage	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
games	.0004132	.002682	0.15	0.878	-.0048679	.0056942
avgmin	.0302278	.0130868	2.31	0.022	.0044597	.055996
points	.0363734	.0150945	2.41	0.017	.0066519	.0660949
rebounds	.0406795	.0229455	1.77	0.077	-.0045007	.0858597
assists	.0003665	.0314393	0.01	0.991	-.0615382	.0622712
_cons	5.648996	.1559075	36.23	0.000	5.34201	5.955982

15. According to the regression results above, what is the F statistic that simultaneously tests the restrictions that all OLS estimators equal to 0.

a) 0.4336
b) 40.27
c) 269
d) 0.0000

Answer: b

16. According to the regression results above, what is the explained sum of squares?

a) 0.6696
b) 90.2698185
c) 117.918945
d) 208.188763

Answer: b

17. In the above regression, which of the variables are significant at the 10% level?

a) *games, avgmin, points, rebounds*
b) *avgmin, points, rebounds*
c) *avgmin, points*
d) *points*

Answer: b

18. In the above regression, which of the variables show a significant **positive** effect at the 1% level?

a) $avgmin, points$
b) $points$
c) $games$
d) None of the variables are positively significant at the 1% level

Answer: b

19. According to the above regression results what is the expected change in wages when $avgmin$ changes, all else remaining constant?

a) Wages will increase by \$30.23 when $avgmin$ increases by 1 unit.
b) Wages will increase by 0.03023% when $avgmin$ increases by 1%.
c) Wages will increase by \$0.03023 when $avgmin$ increases by 1 unit.
d) Wages will increase by 3.023% when $avgmin$ increases by 1 unit.

Answer: d

20. You are considering to use a model with less independent variables:

Source	SS	df	MS
Model	88.1109306	3	29.3703102
Residual	120.077833	265	.453123896
Total	208.188763	268	.776823743

Number of obs	269
F(3, 265)	64.82
Prob > F	0.0000
R-squared	0.4232
Adj R-squared	0.4167
Root MSE	.67314

lwage	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
games	.0001319	.0026829	0.05	0.961	-.0051506	.0054143
avgmin	.0387007	.0097661	3.96	0.000	.0194716	.0579298
points	.0357346	.0151064	2.37	0.019	.0059907	.0654785
_cons	5.650751	.1548824	36.48	0.000	5.345794	5.955707

According to the regression results above, what is the *F*-statistic that simultaneously tests the restrictions that: $\beta_{rebound} = 0$ and $\beta_{assists} = 0$.

a) 2.41
b) 64.82
c) 1.39
d) 3.81

Answer: a

Question 2 (16 points)
Answer parts (a) and (b) below.

(8 points)
(a). Derive the t statistic for the OLS slope coefficient estimator $\widehat{\beta_j}$. State all assumptions required for the derivation.

Error Normality Assumption: The random error terms u_i are independently and identically distributed (iid) as the normal distribution with zero mean and constant variance σ^2
$u_i \sim N(0, \sigma^2)$ **(2 marks)**

Together with the **Gauss-Markov** assumptions, i.e.: **(1 marks)**
MLR.1: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + u$
MLR.2: random sampling from the population
MLR.3: no perfect collinearity in the sample
MLR.4: $E(u|x_1, \dots, x_k) = E(u) = 0$ (exogenous explanatory variables)
MLR.5: $Var(u|x_1, \dots, x_k) = Var(u) = \sigma^2$ (homoskedasticity)

The OLS slope coefficient estimator $\widehat{\beta_j}$ is normally distributed:
$\widehat{\beta_j} \sim Normal[\beta_j, Var(\widehat{\beta_j})]$ **(2 marks)**

The standardized estimators follow a standard normal distribution. **(1 marks)**
$$\frac{\widehat{\beta_j} - \beta_j}{sd(\widehat{\beta_j})} \sim Normal(0, 1)$$

We use $\hat{\sigma}$ as an estimator of σ. Now the standardized estimators follow a t distribution with $n - k - 1$ degree of freedom. The t-statistic for $\widehat{\beta_j}$ thus takes the form: **(2 marks)**
$$t_{\widehat{\beta_j}} = \frac{\widehat{\beta_j} - \beta_j}{se(\widehat{\beta_j})} \sim t_{n-k-1}$$

(8 points)
(b). Explain what is meant by each of the following statements about the estimator $\hat{\beta}$ of the population parameter β.
(1) $\hat{\beta}$ is an unbiased estimator of β.
(2) $\hat{\beta}$ is an efficient estimator of β.

(1) $\hat{\beta}$ is an unbiased estimator of β if **the mean of the sampling (probability) distribution of $\widehat{\boldsymbol{\beta}}$ is equal to $\boldsymbol{\beta}$**, i.e., to the true population value of β:

$\widehat{\boldsymbol{\beta}}$ is an unbiased estimator of $\boldsymbol{\beta}$ if $\boldsymbol{E(\widehat{\beta}) = \beta}$ **(4 marks)**

(2) $\hat{\beta}$ is an efficient estimator of β if it is **unbiased and has smaller variance than any other unbiased estimator** of the parameter β.

If $\tilde{\beta}$ is any other unbiased estimator of β, then $\hat{\beta}$ is an efficient estimator of β if $Var(\hat{\beta}) \leq Var(\tilde{\beta})$ where $E(\hat{\beta}) = \beta$ and $E(\tilde{\beta}) = \beta$ **(4 marks)**

Question 3 (24 points)

You have been commissioned to investigate the relationship between the annual salaries of Chief Executive Officers (CEOs) of firms and the annual profits of their firms. The dependent variable is $\boldsymbol{salary_i}$, the annual salary of the CEO of the i-th firm, measured in *thousands of dollars* per year. The explanatory variable is $\boldsymbol{profits_i}$, the annual profits of the i-th firm, measured in *millions of dollars per year*. The model you propose to estimate is given by the population regression equation

$$salary_i = \beta_0 + \beta_1 profits_i + u_i$$

Your research assistant has used observations on $\boldsymbol{salary_i}$ and $\boldsymbol{profits_i}$ for a sample of 177 corporations to estimate the following OLS sample regression equation, where the figures in parentheses below the coefficient estimates are the *standard errors* of the coefficient estimates:

$$\underset{\quad (45.798)\ (0.10094)}{salary_i = 746.92 + 0.57230 profits_i + \hat{u}_i}$$

(8 points)

(a) Perform a test of the null hypothesis H_0: $\beta_1 = 0$ against the alternative hypothesis H_1: $\beta_1 \neq 0$ at the 1% significance level (i.e., for significance level $\alpha = 0.01$). Show how you calculated the test statistic. State the decision rule you use, and the inference you would draw from the test. Briefly state the conclusion you would draw from the test.

t-test statistic under $\mathbf{H_0}$ is: **(3 marks)**

$$t(\widehat{\beta_1}) = \frac{\widehat{\beta_1} - \beta_1}{se(\widehat{\beta_1})} = \frac{\widehat{\beta_1} - 0}{se(\widehat{\beta_1})} = \frac{0.57230}{0.10094} = \mathbf{5.67}$$

Decision Rule: This is a ***two-tail* test**. Compare the sample t statistic with the α-level critical value of the t_{175} distribution. **(2 mark)**

1. If $|t| \leq c_\alpha$, do not reject H_0 at the significance level α.
2. If $|t| > c_\alpha$, reject H_0 at the significance level α.

Degree of freedom $= n - 2 = 177 - 2 = 175$

Critical value of $\boldsymbol{t_{175}}$ at 1% significance level ($\alpha = 0.01$) is $\boldsymbol{c_{0.01} = 2.576}$ **(1 mark)**

Inference: **(1 mark)**

Since $\boldsymbol{t = 5.67 > 2.576 = c_{0.01}}$**, reject $\mathbf{H_0}$** at the **1%** significance level.

Conclusion: **(1 mark)**

The sample evidence **favors the alternative hypothesis that $\boldsymbol{H_1: \beta_1 \neq 0}$ at the 1% significance level**. It thus provides strong evidence that the annual profits of firms have an impact on the annual salary of their CEO.

(8 points)

(b) Construct the two-sided 95% confidence interval for the slope coefficient β_1

The **two-sided** $(\mathbf{1-\alpha})$ **-level**, or $\mathbf{100(1-\alpha)}$ **percent, confidence interval for** β_1 is computed as

$$\widehat{\beta_1} - c_\alpha se(\widehat{\beta_1}) \leq \beta_1 \leq \widehat{\beta_1} + c_\alpha se(\widehat{\beta_1}) \quad \textbf{(2 marks)}$$

Degree of freedom $= n - 2 = 177 - 2 = 175$
Critical value of $\boldsymbol{t_{175}}$ at 5% significance level ($\alpha = 0.01$) is $\boldsymbol{c_{0.05} = 1.960}$

$$c_\alpha se(\widehat{\beta_1}) = 1.960 * 0.10094 = 0.1978$$

Lower 95% confidence limit for $\boldsymbol{\beta_1}$ **is:** **(3 marks)**

$$\widehat{\beta_1} - c_\alpha se(\widehat{\beta_1}) = 0.5723 - 0.1978 = \mathbf{0.3745}$$

Upper 95% confidence limit for $\boldsymbol{\beta_1}$ **is:** **(3 marks)**

$$\widehat{\beta_1} + c_\alpha se(\widehat{\beta_1}) = 0.5723 + 0.1978 = \mathbf{0.7701}$$

Result: The two-sided 95% confidence interval for $\boldsymbol{\beta_1}$ **is: [0.3745, 0.7701]**

(8 points)
(c) Perform a test of the proposition that an increase in profits of 1 million dollars per year is associated on average with an increase in CEO salary of 500 dollars per year. Use the 5 percent significance level (i.e., $\alpha = 0.05$). State the null hypothesis H_0 and the alternative hypothesis H_1. Show how you calculated the test statistic. State the decision rule you use, and the inference you would draw from the test.

Null hypothesis: $H_0: \beta_1 = 0.5$
Alternative hypothesis: $H_1: \beta_1 \neq 0.5$ a ***two-tail*** **test** **(1 marks)**

t-test statistic under $\mathbf{H_0}$ is: **(3 marks)**

$$t(\widehat{\beta_1}) = \frac{\widehat{\beta_1} - \beta_1}{se(\widehat{\beta_1})} = \frac{\widehat{\beta_1} - 0.5}{se(\widehat{\beta_1})} = \frac{0.57230 - 0.5}{0.10094} = \mathbf{0.716}$$

Decision Rule: This is a ***two-tail*** **test**. Compare the sample t statistic with the α-level critical value of the t_{175} distribution. **(2 marks)**
1. If $|t| \leq c_\alpha$, do not reject H_0 at the significance level α.
2. If $|t| > c_\alpha$reject H_0 at the significance level α.

Degree of freedom $= n - 2 = 177 - 2 = 175$
Critical value of $\boldsymbol{t_{175}}$ at 1% significance level ($\alpha = 0.05$) is $\boldsymbol{c_{0.05} = 1.960}$ **(1 marks)**

Inference: **(1 marks)**
Since $\boldsymbol{t = 0.716 < 1.960 = c_{0.05}}$**, do not reject** $\mathbf{H_0}$ at the **5%** significance level.

Question 4 (30 marks)
You are conducting an econometric investigation into the hourly wage rates of female employees. The sample data consist of 252 observations on the following variables:

$wage_i$ = the hourly wage rate of employee i, measured in dollars per hour;
$educ_i$ = the number of years of formal education completed by employee i;
exp_i = the number of years of work experience accumulated by employee i.

The regression model you propose to use is the log-level regression equation

$$\log(wage_i) = \beta_0 + \beta_1 educ_i + \beta_2 exp_i + \beta_3 educ_i^2 + \beta_4 exp_i^2 + \beta_5 educ_i exp_i + u_i \qquad (1)$$

Using the sample data described above, your research assistant computes OLS estimates of regression equation (1) and of two restricted versions of equation (1). For each of the sample regression equations estimated on the sample of n = 252 observations, the following table contains the OLS coefficient estimates (with estimated standard errors in parentheses below the coefficient estimates) and the summary statistics SSR (residual sum-of-squares), SST (total sum-of-squares), and number of sample observations.

Dependent variable: $\log(wage_i)$				
Independent variables		**(1)**	**(2)**	**(3)**
$educ_i$	β_1	−0.08667 (0.05912)	−0.08817 (0.03822)	0.08231 (0.01046)
exp_i	β_2	0.02102 (0.01461)	0.02059 (0.006428)	0.004116 (0.001894)
$educ_i^2$	β_3	0.007417 (0.002065)	0.007458 (0.001644)	---
exp_i^2	β_4	−0.0003852 (0.0001518)	−0.0003835 (0.000143)	---
$educ_i exp_i$	β_5	−0.000031 (0.0009307)	---	---
$Intercept$	β_0	1.150 (0.4410)	1.162 (0.2399)	0.3348 (0.1415)
SSR		35.4317	35.4319	39.6443
SST		49.5336	49.5336	49.5336
Observations		252	252	252

(10 marks)
(a) Compare the goodness-of-fit to the sample data of the three sample regression equations (1), (2) and (3) in the table. Calculate the value of an appropriate goodness-of-fit measure for each of the sample regression equations (1), (2) and (3) in the table. Which of the three sample regression equations provides the best fit to the sample data? Which of the three sample regression equations provides the worst fit to the sample data?

Must use **adjusted R-squared**, because the regression equations (1) to (3) have **different numbers of regressors**

Eq (1): $\bar{R}^2 = 1 - \frac{SSR/n-k-1}{SST/n-1} = 1 - \frac{35.4317/246}{49.5336/251} = 0.2702$ **(2 marks)**

Eq (2): $\bar{R}^2 = 1 - \frac{SSR/n-k-1}{SST/n-1} = 1 - \frac{35.4319/247}{49.5336/251} = 0.2731$ **(2 marks)**

Eq (3): $\bar{R}^2 = 1 - \frac{SSR/n-k-1}{SST/n-1} = 1 - \frac{39.6443/249}{49.5336/251} = 0.1932$ **(2 marks)**

Eq (2) provides the best fit, while Eq (3) provides the worst fit. **(4 marks)**

(10 marks)

(b) State the coefficient restriction(s) that regression equation (3) imposes on regression equation (1). Explain in words what the restriction(s) mean. Use the results given above from OLS estimation of regression equations (1) and (3) to perform a test of these coefficient restrictions at the 1 percent significance level (i.e., for significance level $\alpha = 0.01$). State the null and alternative hypotheses, show how you calculate the required test statistic. State the decision rule you use, and the inference you would draw from the test. Based on the outcome of the test, which of the two regression equations would you choose, equation (1) or equation (3)?

Equation (3) imposes on equation (1) the exclusion restrictions $\boldsymbol{\beta_3 = 0, \beta_4 = 0}$ **and** $\boldsymbol{\beta_5 = 0}$ **(1 marks)**

Test: $H_0: \beta_3 = 0, \beta_4 = 0, \beta_5 = 0$ versus: H_1: H_0 is not true **(1 marks)**

Interpretation of $\boldsymbol{H_0}$: The quadratic terms of years of education and working experience ($educ_i^2$, exp_i^2) and the interaction term between them ($educ_i exp_i$) do not affect the nature log of hourly wage ($\log(wage_i)$. **(1 marks)**

Compute F statistic: **(3 marks)**

$$F = \frac{(RSS_r - RSS_{ur})/q}{SSR_{ur}/(n-k-1)} = \frac{(39.6443 - 35.4317)/3}{35.4317/246} = 9.7493$$

Decision Rule: Compare the sample F statistic with the α-level critical value of the $\mathbf{F_{3,246}}$ distribution. **(1 marks)**

1. If $F \leq c_\alpha$, do not reject H_0 at the significance level α.
2. If $F > c_\alpha$, reject H_0 at the significance level α.

Critical value of $\mathbf{F_{3,246}}$ at 1% significance level ($\alpha = 0.01$) is $\boldsymbol{c_{0.01} = 3.78}$ **(1 marks)**

Inference: **(1 marks)**

Since $\boldsymbol{F = 9.7493 > 3.78 = c_{0.01}}$**, reject $\mathbf{H_0}$** at the **1%** significance level.

Choose equation (1): restrictions incorporated in equation (2) are rejected at a sufficiently low significance level. **(1 marks)**

(10 marks)

(c) Write the expression for the marginal effect of $educ_i$ on $\log(wage_i)$ implied by regression equation (1). Compute a test of the null hypothesis that the marginal effect of $educ_i$ on $\log(wage_i)$ equals zero for all employees in regression equation (1). Perform the test at the 1 percent significance level (i.e., for significance level $\alpha = 0.01$). State the null hypothesis H_0

and the alternative hypothesis H_1. Write the *restricted* regression equation implied by the null hypothesis H_0. OLS estimation of this *restricted* regression equation yields an **R-squared value** or $\boldsymbol{R^2 = 0.0428}$. Use this information, together with the results from OLS estimation of equation (1), to calculate the required test statistic. State the decision rule you use, and the inference you would draw from the test.

The **marginal effect of** $educ_i$ on $\log(wage_i)$ is:
$\frac{\partial \log(wage_i)}{\partial educ_i} = \beta_1 + 2\beta_3 educ_i + \beta_5 exp_i$ **(2 marks)**

Null hypothesis: $H_0: \beta_1 = 0,\ \beta_3 = 0, \beta_5 = 0$
Alternative hypothesis: H_1: H_0 is not true **(1 marks)**

Restricted model implied by the null hypothesis: **(1 marks)**
$\log(wage_i) = \alpha_1 + \alpha_2 exp_i + \alpha_3 exp_i^2 + u_i$

Find the SSR for the restricted model (SSR_r):
$R_r^2 = 1 - SSR_r/SST$

$SSR_r = (1 - R_r^2)SST = (1 - 0.0428)49.5336 = 47.4136$

Compute F statistic: **(3 marks)**
$$F = \frac{(RSS_r - RSS_{ur})/q}{SSR_{ur}/(n-k-1)} = \frac{(47.4136 - 35.4317)/3}{35.4317/246} = 27.7299$$

Decision Rule: Compare the sample F statistic with the α-level critical value of the $\mathbf{F_{3,246}}$ distribution. **(1 marks)**
1. If $F \leq c_\alpha$, do not reject H_0 at the significance level α.
2. If $F > c_\alpha$, reject H_0 at the significance level α.

Critical value of $\mathbf{F_{3,246}}$ at 1% significance level ($\alpha = 0.01$) is $\boldsymbol{c_{0.01} = 3.78}$ **(1 marks)**

Inference: **(1 marks)**
Since $\boldsymbol{F = 27.7299 > 3.78 = c_{0.01}}$**, reject $\mathbf{H_0}$** at the **1%** significance level.